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A Reconfigurable Auxiliary Circuit for Transient Enhancement in Processor Power Systems with Multiple Voltage Rails

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Post Conference Paper "A Practical Auxiliary Circuit for Voltage Fluctuation Reduction in Multi-core CPU Power Supplies" [19]

***Abstract*—This paper proposes a practical auxiliary circuit for multiple voltage rails to improve dynamic response under significant load steps with an ultrafast slew rate. The proposed auxiliary circuit consists of a current injection/absorption circuit with nonlinear control that can effectively reduce undershoot and overshoot during transients, a sampling mechanism supporting its independent operation, and a reconfigurable energy path supporting multiple voltage rails. The proposed auxiliary circuit is illustrated and verified in a high-power supply system for processors with two voltage rails at different voltage levels, which effectively improves the dynamic response of both voltage rails and demonstrates robustness in fast alternating load steps without affecting regular operation. The maximum load step of 250A with the current slew rate of 1250A/μs is tested, and the overshoot and undershoot are reduced by 36% and 25%, respectively, with recovery times less than 9μs. Compared to existing research and solutions, the proposed auxiliary circuit has cost and space savings advantages, better versatility to support a wide range of power supply types, and is easy to implement.**

***Index Terms*—Auxiliary circuit, dynamic response, nonlinear control, reconfigurable energy path.**

I. Introduction

IN datacenter applications, it is well known that powerful CPUs, GPUs, and AI chips consume more significant amounts of current to meet the exponential growth of artificial intelligence demand for computing power. The processors range from tens to thousands of amperes, often fluctuating rapidly at thousands of amperes per microsecond [1]. To suppress voltage spikes from increasingly aggressive load current steps, years of research on Voltage Regulator modules (VRMs) have introduced nonlinear control for transients, pushing VRM dynamic response to the limits of the LC filter [2]-[5]. Therefore, modifying the LC filter seems the only remaining option, but increasing output capacitance occupies excessive space, and reducing inductance harms steady-state efficiency.

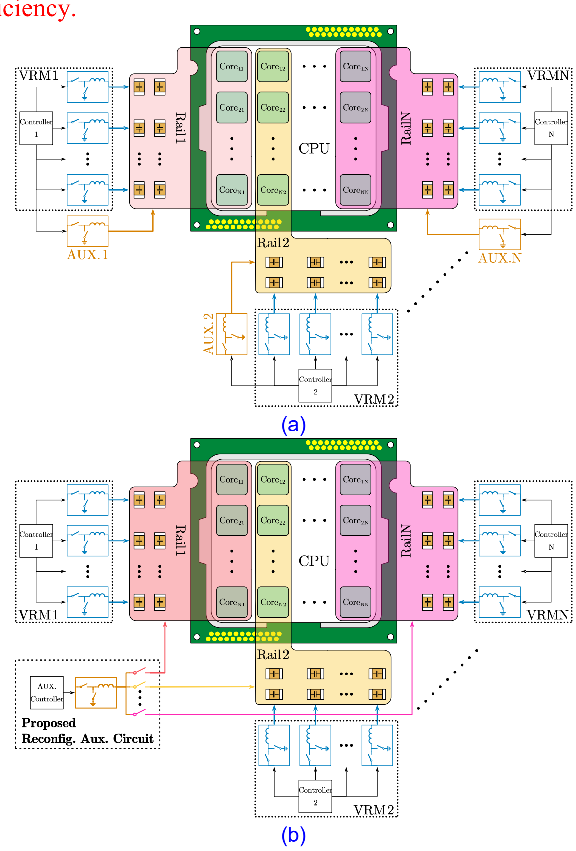

Fig. 1. (a) Conventional Auxiliary circuits. (b) Proposed Auxiliary circuit.

Various auxiliary circuits have been developed to improve the dynamic response without further altering the LC filter [6]-[16]. The general principle is that the auxiliary circuit remains inactive during steady state and quickly injects or absorbs current during transients to compensate for the charge on the output capacitor, reducing voltage spikes while maintaining high steady-state efficiency. Due to differences in control strategies, implementations, and application scenarios,

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each solution offers distinct advantages and disadvantages. In [6]-[8], linear regulators are used as auxiliary circuits, offering faster response due to the absence of components like inductors, but they suffer from significant energy losses and are limited to managing small current steps. In [9], an additional switching converter triggered by the output capacitor current is proposed. This auxiliary circuit is simple to implement and responsive, but accurately sensing current is challenging due to the large parallel capacitance in modern processor power supplies. In [10], A fully digital current-programmed mode buck converter with an auxiliary phase is proposed, achieving near-ideal transient response through careful analysis and control during transients. However, its reliance on tight coordination between the main converter and auxiliary phase and lookup tables increases complexity when applied to high-power systems with complex protocols. A novel soft-switching control method for the auxiliary circuit is introduced in [11] to reduce losses. However, this approach is not optimal for enhancing dynamic response, and its complex control hinders practical implementation. The Energy Recycling Output Impedance Correction Circuit (OICC) in [12] and [13] supports adaptive voltage position (AVP) and dynamic voltage and frequency scaling (DVFS) features required for processor power supplies. However, its current-mode control method increases voltage ripples during transients, and implementing DVFS requires modifications to the main converter or processor protocols. In [14], a dual-loop control algorithm assigns voltage regulation to auxiliary phases and power transmission to main phases. This architecture naturally satisfies functions such as AVP and DVFS but requires the auxiliary phase to operate continuously, impacting the steady state efficiency. The same problem exists in [15], although it uses a more conventional control algorithm. In [16], a hypothesis was proposed where processors could notify load steps in advance, enabling the auxiliary circuit to prepare and achieve near-zero spikes during transients. However, there is still no indication that this concept will be implemented in the foreseeable future.

Based on the above analysis of previous studies, this paper argues that for processors used in data centers, a practical auxiliary circuit should have the following features:

1) **The effectiveness in improving dynamic response.**

For modern microprocessors, auxiliary circuits must absorb or inject massive current to effectively suppress voltage spikes. Minimizing energy loss in these circuits is essential; otherwise, overall power consumption will suffer, and substantial energy loss will generate heat, raising the risk of thermal failure. Therefore, the auxiliary circuit should utilize switching converters rather than linear regulators.

2) **Compatibility with various VRMs.**

There are a variety of VRMs on the market today with different architectures and control methods. Therefore, the ideal auxiliary circuit should behave like a plug-and-play device that can be used with different VRM solutions. This requires the auxiliary circuit to operate independently and collect the information itself, rather than getting it from the VRM or processor, which requires additional modifications.

3) **Shareability across multiple voltage rails.**

Motherboards often host multi-core processors for data centers, resulting in multiple voltage rails, each demanding substantial current. All currently known auxiliary circuits support only one voltage rail, requiring multiple circuits for multi-rail systems, as shown in Fig.1 (a). In contrast, an auxiliary circuit capable of dynamically adjusting its energy path to assist multiple rails, as shown in Fig. 1(b), offers significant cost and energy efficiency benefits.

4) **Fulfillment of processors' power supply functions.**

Modern processor power supply systems are complex, requiring support for the Power Management Bus (PMBus) and its related protocols, which dynamically adjust parameters like voltage, current, and frequency during operation. Many studies overlook these practical considerations, limiting their applicability to simple, low-power Point-of-Load (PoL) converters, where modifying LC filters is often more practical than deploying an auxiliary circuit.

This paper proposes a practical auxiliary circuit for suppressing voltage spikes on multiple voltage rails that satisfies the features above. The circuit uses a switching converter with nonlinear control to inject or absorb large currents quickly. It operates independently by detecting the state of multiple rails without requiring modifications to the VRMs or processors. Additionally, it can flexibly connect to any voltage rail via reconfigurable energy pathways and employs customized control strategies to adapt to voltage rails with complex protocols.

The rest of this paper is organized as follows: Section II introduces the concepts and scheme of the proposed auxiliary circuit. Section III details its specific implementation. Section IV presents the simulation and experimental results. Finally, Section V provides the conclusion.

II. Proposed Practical Auxiliary Circuit Scheme

The structure of the proposed auxiliary circuit is shown in Fig. 2. The auxiliary circuit contains a sink/source current circuit, sample circuits, several back-to-back MOSFETs, and a nonlinear controller using collected information for operation. The auxiliary circuit can reduce voltage spikes by compensating the charge imbalance by quickly absorbing or injecting current when significant load steps occur.

The basic principle of the proposed auxiliary circuit operation is that the circuit will be activated only when a significant load step occurs. The auxiliary circuit does not involve small load steps that only produce tolerable voltage spikes, so it does not affect the steady-state efficiency of the VRMs. The auxiliary circuit can change the energy path between multiple voltage rails under different voltage rails using back-to-back MOSFETs, but this also means that the circuit can only assist one VRM at a time. This assumption—that significant load steps do not occur simultaneously across multiple rails—is reasonable for CPU power supplies, as technologies like Intel Turbo Boost prevent such events. For other kinds of load, such as GPUs and AI chips, minor firmware and OS modifications can adapt the auxiliary circuit for broader applications.

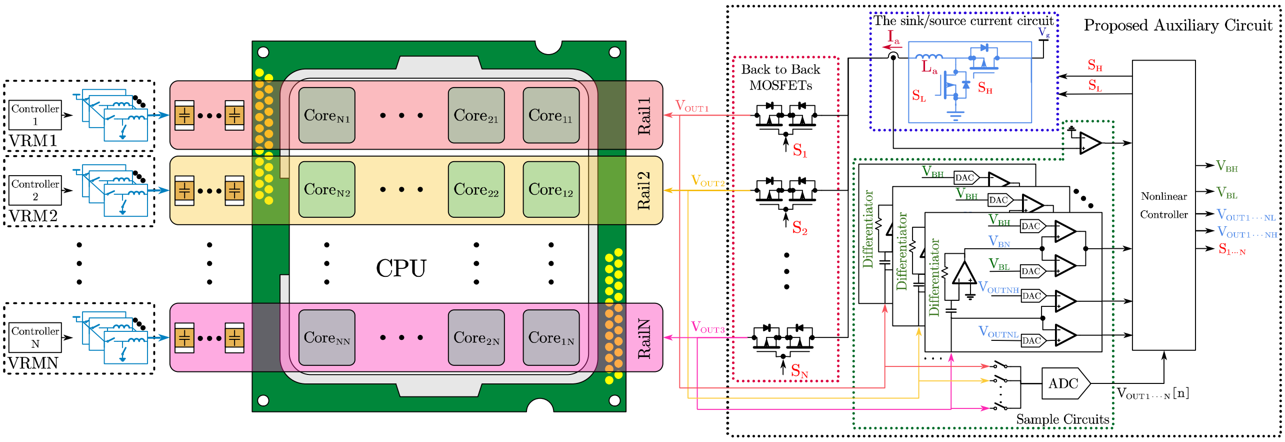

Fig. 2. The architecture of VRMs with the proposed auxiliary circuit.

To make the auxiliary circuit practical, a quantitative analysis of the relationship between the characteristics of the auxiliary circuit and the amount of improvement in voltage spike proceeds to help set the auxiliary circuit. The analysis will proceed for both load step-up and step-down scenarios in the following.

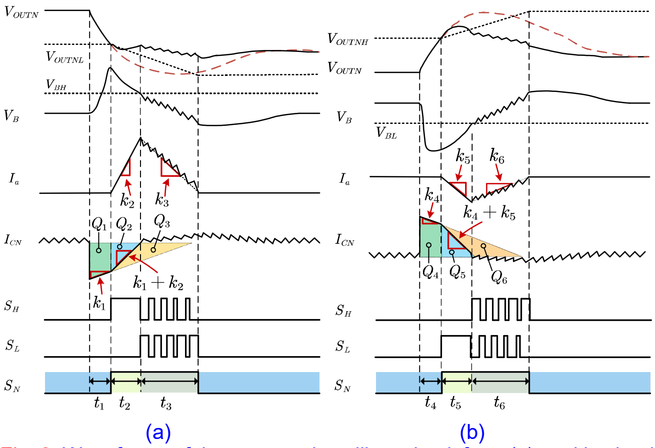

Fig. 3. Waveforms of the proposed auxiliary circuit for a (a) positive load step. (b) negative load step.

A. Load step-up scenarios

As shown in Fig. 3 (a), during a significant load step-up event, the auxiliary circuit injects current quickly to improve dynamic response. The operation of the auxiliary circuit occurs over three periods:

Period t_1: When a light-to-heavy load transient occurs on Rail N, the output voltage V_{OUTN} drops because the VRM is supplying less current than the load demands. The VRM begins increasing its output current to compensate. During this period, the auxiliary circuit remains inactive since the output voltage has not yet reached the threshold V_{OUTNL}.

Period t_2: Once V_{OUTN} reaches the threshold V_{OUTNL}, the auxiliary circuit activates. The auxiliary circuit connects to voltage rail N via corresponding back-to-back MOSFETs and triggers the Buck circuit to inject current into the load. Deciding when this period should end is vital to the overall operation process. The auxiliary circuit should stop boosting the current it injects into the load when the output voltage drops to its minimum value, as the undershoot magnitude stabilizes and will not worsen. At this point, the rate of change of the output voltage (dV_{OUT}/dt) shifts from negative to positive, meaning this period should end when dV_{OUT}/dt becomes zero.

To detect this point, the auxiliary circuit uses a differentiator to obtain the change rate of the output voltage, dV_{OUT}/dt, as detailed later in Section III. In theory, the end of this period occurs when the differentiator's output, V_B, reaches zero. However, in this design, the auxiliary circuit determines when to exit this period by comparing V_B to a threshold value V_{BH}. This approach accounts for the fact that the output capacitance of a high-performance CPU typically consists of multiple arrays of multilayer ceramic capacitors (MLCCs) connected in parallel, leading to complex impedance characteristics in its power transfer network (PDN). A simple differentiator may not fully capture the output voltage slope. For practical and cost-effective reasons, an adjustable threshold is employed to pinpoint when the output voltage reaches its minimum, signaling the end of this period for the auxiliary circuit.

Period t_3: The sink/source current circuit will reduce the amount of injected current during this period. The rate at which the injected current decreases is carefully controlled to ensure it remains within an appropriate range. This allows the VRM to ramp up its output current in time, preventing any further voltage spikes. Once the injected current reaches zero, the back-to-back MOSFETs disconnect, signaling the end of the auxiliary circuit's operation, which then awaits the next activation.

From the operation of the auxiliary circuit, it is evident that analyzing the rate of injected current is essential for improving the dynamic response and avoiding extra voltage spikes. Considering the characteristics of the CPU power supply, the voltage drop caused by the equivalent series resistance (ESR) is neglected because of MLCC arrays that have relatively low ESR. Similarly, the rate of load steps is neglected since it is much faster than the VRM's response time.

In most modern VRM solutions, regardless of the manufacturer, the controller typically switches to a nonlinear control method during large transients to improve dynamic response. For instance, a VRM utilizing Constant-on-time (COT) control will output PWM signals at fixed intervals, while a VRM using PID control will output PWM signals at a constant duty cycle during this period. Regardless of the control method used, the rate of current rise to the load remains constant due to the fixed PWM signals. **The rate of current increase with VRM alone is defined as k_1.**

Based on the property that the rate of current rise is constant, we can calculate the magnitude of output voltage drop ΔV_{drop} for a given current step when using the VRM alone. This voltage drop occurs due to the gap between the output current and the load current during the transient, with the output capacitor discharging to compensate for the load. The shaded triangular area $(Q_1+Q_2+Q_3)$ of the output capacitor current waveform in Fig. 3 (a) represents the charge supplied by the capacitor to the load during the transient. By considering the relationship between the current and voltage for a capacitor $I=\mathrm{d}Q/\mathrm{d}t=C\cdot \mathrm{d}V/\mathrm{d}t$, the voltage drop ΔV_{drop} resulting from a large current step ΔI handled by the VRM only can be derived as:

$$\Delta V_{drop}=\frac{Q_1+Q_2+Q_3}{C_{OUT}}=\frac{\Delta I^2}{2C_{OUT}\cdot k_1} \tag{1}$$

Next, we calculate how much the voltage drop during a step-up transient can be mitigated with the assistance of the auxiliary circuit. The auxiliary circuit remains idle until the output voltage reaches the undershoot detection threshold V_{OUTNL}. During this period, the characteristics of the output voltage drop are the same as if the VRM were operating alone. Therefore, the period t_1 from the occurrence of the transient to the activation of the auxiliary circuit can be calculated using the relationship between charge and voltage for capacitors $\Delta Q=C\cdot\Delta V$, as shown in the following equation:

$$\Delta V_{OUTLN}=V_{OUTN}-V_{OUTLN}=\frac{Q_1}{C_{OUT}}=\frac{(2\Delta I-k_1t_1)t_1}{2C_{OUT}} \tag{2}$$

And t_1 can be derived as:

$$t_1=\frac{\Delta I-\sqrt{\Delta I^2-2k_1\Delta V_{OUTLN}C_{OUT}}}{k_1} \tag{3}$$

Once the output voltage falls below the threshold V_{OUTNL}, the auxiliary circuit begins injecting current into the load. During the period t_2, from the activation of the auxiliary circuit until the output voltage reaches its minimum, the auxiliary circuit injects current at a constant rate, **which can be defined by the current injection slope k_2.**

With the assistance of the auxiliary circuit, the rate of current injection to the output capacitor is increased from k_1 (when only the VRM is operating) to k_1+k_2 (when the VRM and auxiliary circuits are operating together). As a result, the charge supplied by the output capacitor to the load is reduced to the area indicated by Q_1+Q_2 in the Fig. 3 (a), which consequently decreases the output voltage drop. Using the relationship between charge and voltage for a capacitor, the improved voltage drop ΔV_{drop_imp} can be derived as:

$$\Delta V_{drop\_imp}=\frac{Q_1+Q_2}{C_{OUT}}=\frac{1}{k_1+k_2}\frac{\Delta I^2}{2C_{OUT}}+\frac{k_2}{k_1+k_2}\Delta V_{OUTLN} \tag{4}$$

Using (4), the desired value of k_2 can be determined based on the target output voltage drop. The appropriate value of k_2 for reducing the undershoot can be derived as:

$$k_2=\frac{1-\frac{\Delta V_{drop\_imp}}{\Delta V_{drop}}}{\Delta V_{drop\_imp}-\Delta V_{OUTLN}}\frac{\Delta I^2}{2C_{OUT}} \tag{5}$$

According to (5), the current injection rate k_2 for the sink/source current circuit can be set based on the maximum possible current step $\Delta I_{\max}$, the desired maximum undershoot voltage ΔV_{drop_imp}, and output capacitance C_{OUT}. This equation provides a guide for easily deploying the auxiliary circuit.

During the period of t_3, after the charge on the output capacitor has been compensated and the output voltage no longer drops, the sink/source current circuit will gradually reduce the injected current, allowing the auxiliary circuit to disengage smoothly. Simultaneously, the VRM will increase its output current to make up for the reduced auxiliary current. To prevent the auxiliary circuit from reducing the current too quickly, which could cause another voltage undershoot, the auxiliary circuit alternates switches S_H and S_L with a set duty cycle d_1 during t_3 to control the rate of current reduction. **The rate of current reduction under the set duty cycle by the auxiliary circuit is defined as k_3**, which can be derived as:

$$k_3=\frac{V_gd_1-V_{OUTN}}{L_a} \tag{6}$$

The value of k_3 should be carefully controlled to avoid disturbing the output voltage. Theoretically, k_3 should be equal to k_1, ensuring the auxiliary circuit reduces the current at the same rate as the VRM and does not cause another undershoot. In practice, it is recommended that k_3 be set slightly lower than k_1, with $k_3\leq 2/3\cdot k_1$, to ensure a smooth disengagement of the auxiliary circuit.

B. Load step-down scenarios

The waveform with the assistance of the auxiliary circuit during a significant load step-down event is shown in Fig. 3 (b). The auxiliary circuit operates similarly to the step-up scenario:

Period t_4: A heavy to light transient occurs on Rail N, causing the output voltage V_{OUTN} to rise as the VRM supplies more current than the load requires. The VRM begins to reduce its output current to drain the excess charge, while the auxiliary circuit remains idle until the output voltage reaches the threshold V_{OUTNH}.

Period t_5: Once V_{OUTN} reaches the threshold V_{OUTNH}, the auxiliary circuit activates by closing the back-to-back MOSFETs and absorbing current from the load. This period ends when the V_B reaches the threshold value V_{BL}, indicating that the output voltage has peaked during the transient.

Period t_6: The sink/source current circuit gradually reduces

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the absorbed current at a controlled rate until it reaches zero. At this point, the back-to-back MOSFETs disconnect, completing the operation.

A similar analysis can also be applied to calculate the reduction in overshoot. The overshoot, like the undershoot, results from the mismatch between the load current and the VRM output current, except in this case, the output current exceeds the load requirement, leading to an accumulation of excess charge on the output capacitor and thus an overshoot. Similarly, the area of the shaded triangle ($Q_4+Q_5+Q_6$) of the output capacitor current waveform in Fig. 3 (b) represents the charge accumulated to the load during the transient. The voltage rise ΔV_{over}, caused by a relatively large current step ΔI handled solely by the VRM, can be derived as:

$$\Delta V_{over}=\frac{Q_4+Q_5+Q_6}{C_{OUT}}=\frac{\Delta I^2}{2C_{OUT}\cdot k_4} \tag{7}$$

Where k_4 is the rate at which the output current decreases when only the VRM is operating.

The period t_4, from the transient occurrence to the triggering of the auxiliary circuit, can be expressed as:

$$\Delta V_{OUTHN}=V_{OUTN}-V_{OUTHN}=\frac{Q_4}{C_{OUT}}=\frac{(2\Delta I-k_4t_4)t_4}{2C_{OUT}} \tag{8}$$

The value of t_4 can then be derived as:

$$t_4=\frac{\Delta I-\sqrt{\Delta I^2-2k_4\Delta V_{OUTHN}C_{OUT}}}{k_4} \tag{9}$$

Once the threshold is triggered, the auxiliary circuit begins sourcing current from the output during t_5, and **the rate of current absorption by the auxiliary circuit is defined as k_5.** The combined rate of current absorption from the output increases to k_4+k_5, which reduces the accumulated charge on the output capacitor to Q_4+Q_5. The improved voltage overshoot ΔV_{over_imp} and the value of k_5 can be derived as:

$$\Delta V_{over\_imp}=\frac{Q_4+Q_5}{C_{OUT}}=\frac{1}{k_4+k_5}\frac{\Delta I_{\max}^2}{2C_{OUT}}+\frac{k_5}{k_4+k_5}\Delta V_{OUTHN} \tag{10}$$

$$k_5=\frac{1-\frac{\Delta V_{over\_imp}}{\Delta V_{over}}}{\Delta V_{over\_imp}-\Delta V_{OUTHN}}\frac{\Delta I^2}{2C_{OUT}} \tag{11}$$

Similar to the load step-up scenarios, during t_6, the auxiliary circuit switches S_H and S_L with a set duty cycle d_2 to control current reduction rate. **The rate of current reduction with this duty cycle by the auxiliary circuit is defined as k_6,** which can be derived as:

$$k_6=\frac{V_gd_2-V_{OUTN}}{L_a} \tag{12}$$

k_6 should be carefully controlled to ensure the VRM reduces the current smoothly, avoiding subsequent voltage spikes. In practice, k_6 should satisfy $k_6 \leqslant 2/3\cdot k_4$ to achieve a smooth transition.

III. IMPLEMENTATION

The implementation of the proposed auxiliary circuit in this paper has been carefully considered to satisfy the features mentioned in the introduction section.

A. Effectiveness in improving dynamic response

To effectively suppress voltage spikes, the sink/source circuit needs to be able to rapidly inject or absorb large amounts of current into the load. In the proposed auxiliary circuit, a single-phase buck converter is used, but any other topology will also work as long as it can act as a controlled current source. The single-phase Buck converter consists of switches S_H and S_L, along with an inductor L_a with a relatively small inductance value. The input of the single-phase buck converter is the same as the VRMs, and with the help of the small inductor and nonlinear control method, the circuit can fulfill the requirement to reduce voltage spikes effectively. During load step-up, S_H turns on to inject current into the load and suppress undershoot, while during load step-down, S_L turns on to absorb current from the load and suppress overshoot. Since the auxiliary circuit functions only during transients, the inductance of L_a can be minimized to enhance the voltage spike without affecting the steady-state efficiency of the VRM.

For this specific design, the rate of injection and absorption of current is mainly related to the inductance of L_a and the duty cycle. The value of k_2, k_3, k_5, and k_6 in the previous analysis can be derived as:

$$\begin{cases} k_2=\frac{V_g-V_{OUTN}}{L_a} & k_3=\frac{V_gd_1-V_{OUTN}}{L_a} \\ k_5=\frac{V_{OUTN}}{L_a} & k_6=\frac{V_gd_2-V_{OUTN}}{L_a} \end{cases} \tag{13}$$

Where V_g is the input voltage of the auxiliary circuit. d_1 and d_2 is the duty cycle of the single-phase buck converter during the injected current reduction and absorbed current reduction, respectively. The value of L_a can then be derived according to (13) as:

$$L_a=\begin{cases} \frac{\Delta V_{drop\_imp}-\Delta V_{OUTLN}}{1-\frac{\Delta V_{drop\_imp}}{\Delta V_{under}}}\frac{2C_{OUT}(V_g-V_{OUTN})}{\Delta I_{\max}^2}, & according\ to\ k_2 \\ \frac{\Delta V_{over\_imp}-\Delta V_{OUTHN}}{1-\frac{\Delta V_{over\_imp}}{\Delta V_{over}}}\frac{2C_{OUT}V_{OUTN}}{\Delta I_{\max}^2}, & according\ to\ k_5 \end{cases} \tag{14}$$

Where $\Delta I_{\max}$ is the maximum possible current step of the load. The inductance value of L_a will be finally determined depending on the magnitude of the overshoot and the undershoot. For this specific application where the mainstream VRMs are typical multiphase interleaved Buck converters to convert an input of 6~12V to an output around 1V, these VRMs' ability to inject current is much better than the ability to absorb current, usually resulting in that overshoot is much more significant than undershoot, so in this design the inductance value is selected with reference to k_5.

B. Compatibility with various VRMs

The timing and manner of operation of the auxiliary circuit are subject to its own information gathering and judgment to allow it to be compatible with the currently available VRM products. The adjustable thresholds V_{OUTNL} and V_{OUTNH} are output through digital-to-analog converters (DACs), and the

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determination process is realized through a window comparator. The point at which the auxiliary circuit ceases to absorb or inject current is determined by a differentiator circuit designed using the impedance matching methodology outlined in [17]. The output voltage of this circuit V_B is related to the derivative of the output voltage ($\mathrm{d}V_{OUT}/\mathrm{d}t$).

However, in processor power supplies that typically contain numerous parallel capacitors and vias on a large power plane, the complexity of the power distribution network (PDN)—which may involve thousands of parasitic parameters—can lead to inaccuracies in the differentiator's output. To address this inaccuracy, rather than attempting to resolve the complexities of the PDN, we employ a window comparator and DACs to introduce offsets V_{BH} and V_{BL} to compensate for the error made by PDN. This approach provides greater flexibility and simplicity in practical applications.

Ideally, maintaining a constant duty cycle according to (13) in the period of reduction, the injected/absorbed current will reduce the current to 0 at a constant rate. However, as shown in Fig. 4 (a), due to the presence of parasitic parameters such as the inductive dc resistance (DCR) and the current sampling resistors, the rate is not constant at higher currents. The more accurate rate can be derived as:

$$\begin{cases} k_3' = \dfrac{V_g d_1 - V_{OUTN}}{L_a} - \dfrac{R_{eq}}{L_a} I_a \\ k_6' = \dfrac{V_g d_2 - V_{OUTN}}{L_a} - \dfrac{R_{eq}}{L_a} I_a \end{cases} \tag{15}$$

Where R_{eq} is the equivalent resistance, including DCR, current sample resistor, drain-source on resistance of a MOSFET ($R_{DS(on)}$), etc. This leads to a situation where, in practice, the current reduces at a desired rate at first, but then the rate slows down as the current reduces, which will cause the auxiliary circuit to take a long time to wait for the current to return to zero and disconnect from the load. The ideal solution to this problem would be to have the duty cycle of each cycle adjusted to the value of the current I_a. However, from a more practical point of view, as shown in Fig. 4 (b), the solution used in this design is to use the desired duty cycle for the first three cycles and then change to a smaller duty cycle after that. Also, the fact that the rate slows down as the current reduces makes the auxiliary circuit disconnect more smoothly.

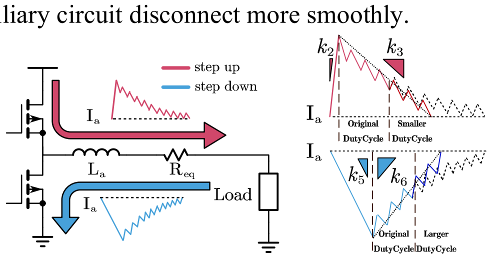

Fig. 4. (a) non-ideal auxiliary current with parasitic resistance. (b) Current slope correction by duty cycle changes.

C. Shareability across multiple voltage rails

In order to share between various VRMs that operate in different voltage rails, the auxiliary circuit needs to be able to reconfigure its energy transfer path. The reconfigurable energy path should be able to let the current flow in both directions, completely isolate between energy paths, and minimize the switching and conduction losses.

To meet these requirements, back-to-back MOSFETs are used in this design to allow current to flow in both directions when closed and completely isolate different voltage rails when disconnected. In addition, since the CPU power supply voltage generally does not exceed 2V, it is possible to use MOSFETs with low drain-source breakdown voltage which have small $R_{DS(on)}$ and low-side drivers, which are good for reducing losses and costs, respectively.

D. Fulfillment of processors' power supply functions

Accurately recognizing transients and distinguishing them from other voltage-changing functions is crucial for practical implementation. Two key functions that can interfere with the auxiliary circuit and require additional measures are dynamic voltage and frequency scaling (DVFS) and adaptive voltage positioning (AVP).

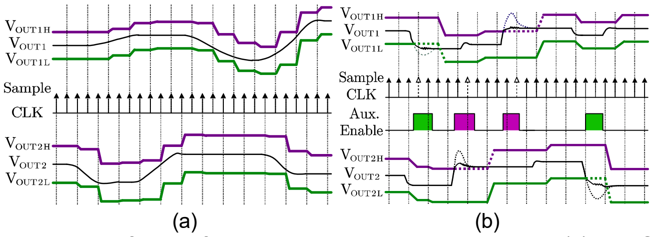

Fig. 5. Waveforms of output voltage and thresholds with (a) DVFS function and (b) AVP function.

The DVFS function is a widely used feature in CPUs that adjusts the voltage according to clock frequency variations to save energy. Voltage adjustment is realized according to the instructions given by the CPU to the VRM at a rate typically in the range of a few millivolts per microsecond. In this design, as shown in Fig. 5 (a), an analog-to-digital converter (ADC) is used to sample the output voltage V_{OUTN} and dynamically adjust the thresholds V_{OUTNL} and V_{OUTNH} output by the DACs based on the sampled voltage value. Due to the slow voltage changes under DVFS, it is possible to sample multiple output voltages in turn with a single low sample rate ADC, which is a cost-effective solution.

The AVP function is also a popular feature for reducing voltage spikes during the transient. The basic idea of AVP is to have a higher voltage level at light load and a lower voltage level at full load so that the overall magnitude of voltage spikes can be reduced to within the voltage tolerance window. In order for the auxiliary circuit to be used with the AVP function when a transient occurs, the ADC mentioned in the DVFS function samples the output voltage after the auxiliary circuit completes its operation and adjusts the thresholds V_{OUTNL} and V_{OUTNH} according to the sampled value, as shown in Fig. 5 (b). During the operation of the auxiliary circuit, the threshold signal will be ignored to prevent false triggering.

Simulations were conducted using SIMPLIS with the specifications as shown in Table I. The features and performance are verified, as shown in Fig. 6. The output voltages of the two Rails, V_{OUT1} and V_{OUT2}, were set at 1 V and 0.88 V, respectively. In the first step, the current in Rail 1 was ramped up from 0 A to 250 A at a rate of 1250 A/µs, and after 20 µs, the current in Rail 2 was reduced from 250 A to 0 A at the same rate. The simulation results clearly show that with the assistance of the auxiliary circuits, the undershoot in V_{OUT1} was reduced from 60 mV to 29 mV, and the overshoot in V_{OUT2} was reduced from 135 mV to 65 mV. This demonstrates the effectiveness of the auxiliary circuit in suppressing both the undershoot and overshoot of the output voltages. Moreover, by observing the inductor current waveforms of the auxiliary circuit I_a, it is evident that the current slope correction successfully maintains a consistent slope during the period of reduction of the auxiliary circuit, which prevents the current reduction from slowing down excessively. This correction shortens the overall operation time of the auxiliary circuit compared to the case where the correction is not applied, thereby improving recovery time. The simulation results validate the effectiveness of the proposed auxiliary circuit in enhancing both undershoot and overshoot suppression, providing a solid foundation for the experimental verifications in the next section.

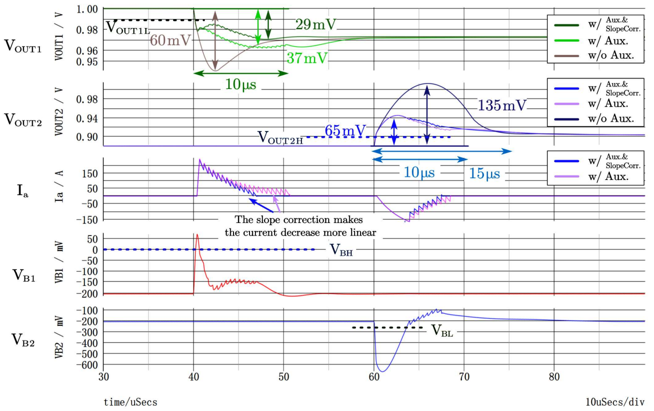

Fig. 6. Waveforms of both voltage rails with transients under simulation.

IV. Experimental Results

A full-scale prototype with the specifications in Table I is established to verify the functions and performance of the proposed auxiliary circuit, as shown in Fig. 7.

The VRMs consist of two 6-phase Buck converters utilizing current-mode constant-on-time (CMCOT) control based on a pulse distribution method, which is widely recognized and commonly employed in products from Monolithic Power Systems (MPS). As shown in Fig. 8, the system compares the difference between the output voltage and the reference voltage, along with a slope for stability compensation, against the total inductor current using a comparator. This comparison generates timing pulses that determine when each phase is activated. The pulses are directed to a constant-on-time generator via a phase manager, producing the PWM signals required for normal operation. The specific design features a hybrid digital-analog architecture, where the current sampling, voltage sampling, slope compensation and comparison in this prototype are built from analog circuits, and the generated pulse signals are passed through the FPGA for phase assignment and constant on-time generation, and the generated PWM signals are passed to the DrMOS from MPS (MP86957).

The auxiliary circuit is controlled by a Xilinx Kintex-7 FPGA and the sink/source current circuit is implemented by three DrMOS (MP86957) with 36nH inductance each, controlled by the same signal, which can be equivalently treated as a single Buck converter with 12nH inductance. The Back-to-Back MOSFETs are implemented by low-$R_{DS(on)}$ MOSFETs (IPB200N25N3G) and low-side gate drivers (2EDN7524G). The DACs are implemented by LTC2630, and the ADC is implemented by LTC2297. Although the LTC2297 can achieve 40Msps synchronous sampling, the controller only accepts incoming data at 1Msps in turn, which means it can be replaced with a much cheaper product. All parts of the auxiliary circuit are implemented by discrete devices and take up some area now, which can be greatly reduced if they are integrated in the future.

TABLE I
System Specifications

Specifications	Values	Remark
VRMs (Two 6 phases COT Buck converters)		
Input voltage V_g	12V	
Output voltage V_{OUTN}	Rail1: 1.0V Rail2: 0.8V	
AVP parameter R_i	0.1mΩ	ΔV=I×R_i
VRM Inductor L_n	150nH	
Output capacitance C_{OUT}	6.3mF	134×47µF
Switching frequency f_{SW}	1MHz	
Auxiliary Circuit		
Inductor L_a^*	12nH	The equivalent value of three 36nH inductors
Positive threshold V_{OUTNL}	25mV	
Negative threshold V_{OUTNH}	30mV	

*: Three 36nH Buck converters with identical controls are used to equate a single 12nH Buck converter due to a shortage of inductor inventory

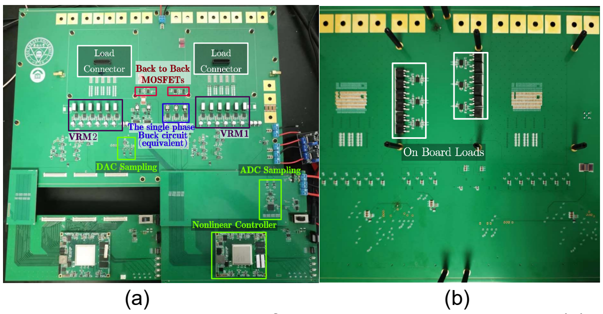

Fig. 7. Experimental prototype of the proposed auxiliary circuit. (a) Top side. (b) Bottom side.

The transient response is tested by LoadSlammer Pro 1000 from ProGrAnalog, which can perform maximum 250A load step tests with a slew rate of 1250A/µs. The waveforms are presented with the load steps of 50-300A and 300-50A in Fig. 9. The auxiliary circuit effectively suppresses the overshoot by

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36% and undershoot by 25%.

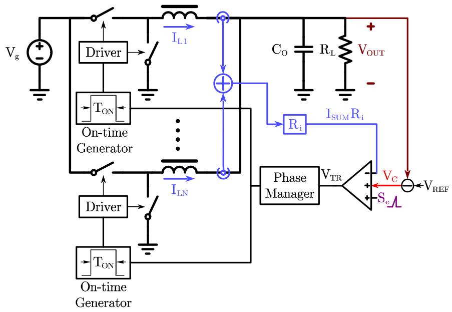

Fig. 8. The schematic diagram of the VRM1,2 using COT control based on pulse distribution method.

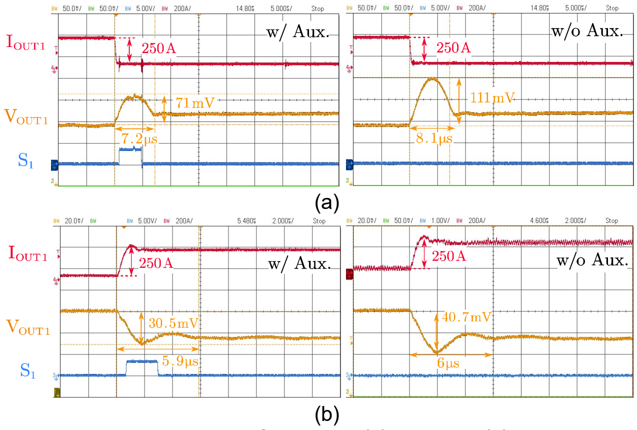

Fig. 9. Transient response of ΔI=250A. (a) 300-50A. (b) 50-300A.

Fig. 10 shows the transient performance of the VRM with and without the help of the auxiliary circuit under other cases of load steps, further demonstrating the effectiveness of the auxiliary circuit. There is no waveform of the auxiliary circuit to improve the 50-250A and 150A-300A situations, mainly because the undershoot in these cases does not exceed 25mV and, therefore, does not trigger the auxiliary circuit, which is reasonable and in line with the expectation that small load steps will be handled by the VRM alone.

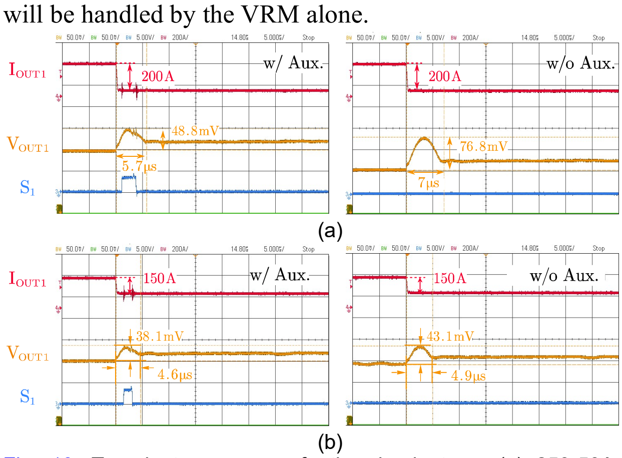

Fig. 10. Transient response of other load steps. (a) 250-50A. (b) 300-150A.

Tests of both voltage rails are realized by on-board loads shown in Fig. 7 (b). The on-board load comprises MOSFETs (IQE006NE2LM5) directly connected to the voltage rails and ground. In these tests, the amplitude of positive load step-up and negative load step-down is around 450A and 250A, respectively. The current slew rate of the on-board load is faster than the LoadSlammer due to parasitic parameters on the load connector, so these tests can be used to demonstrate the effectiveness and performance of the auxiliary circuit.

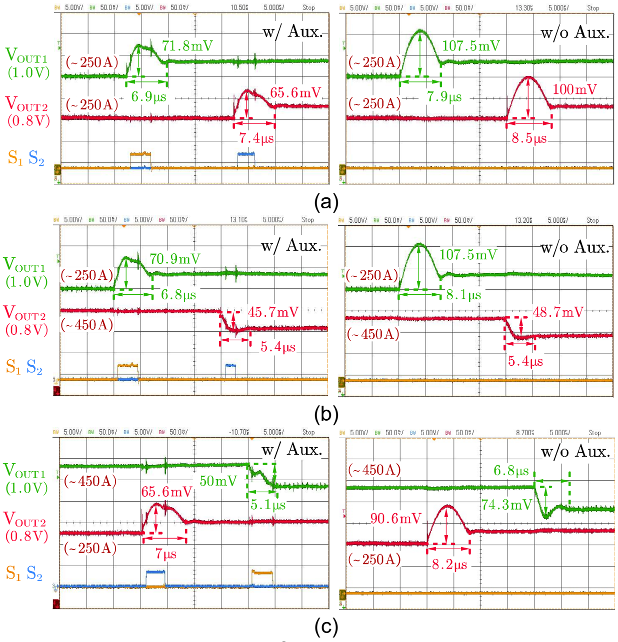

Fig. 11. Transient response of load steps occurs in both voltage rails with an interval of 20μs. (a) VOUT1 overshoot & VOUT2 overshoot. (b) VOUT1 overshoot & VOUT2 undershoot. (c) VOUT1 undershoot & VOUT2 overshoot.

Fig. 11 shows the improvement of the dynamic response of the auxiliary circuit for both voltage rails. The auxiliary circuit can quickly and efficiently assist both voltage rails with different output voltages with a 20μs interval between load steps, confirming the feasibility of the idea that the proposed auxiliary circuits can be shared between voltage rails.

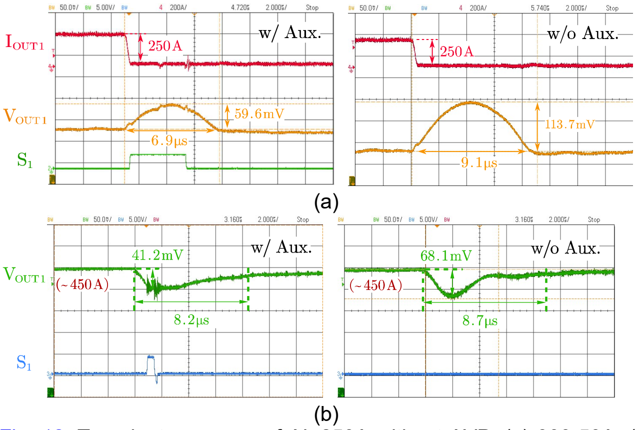

Fig. 12. Transient response of ΔI=250A without AVP. (a) 300-50A. (b) 0-~450A.

Fig. 12 shows the assistance of the auxiliary circuit for voltage rails that do not use the AVP function. The overshoot under a 250A load step and the undershoot under a 450A load step are reduced by 48% and 40%, respectively, demonstrating that the auxiliary circuit can be used in various power supply

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types, including VRMs and PoLs. What's more, in PoL applications such as GPU and AI chips, the ADC can be removed because its output voltage is not adjusted in real time, making it cheaper to deploy the proposed auxiliary circuit.

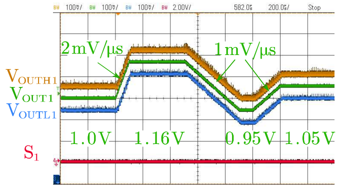

Fig. 13. Waveforms of output voltage and thresholds with DVFS.

Fig. 13 presents the adjustments of thresholds with the DVFS function. The output voltage is adjusted to multiple values at different slew rates according to the commands, and the thresholds follow in time to avoid interference with the VRM caused by a false triggering of the auxiliary circuit.

The results of the auxiliary circuit's suppression of voltage spikes are summarized in Table II, and the comparison with several conventional VRMs and previous auxiliary circuit solutions is given in Table III. A figure of merit (FoM) that has been proposed in the [7] is used in the comparison to evaluate the output capacitance requirement of each design. The proposed solution has a much higher FoM compared to others, indicating a much lower output capacitance requirement and a great space-saving potential.

TABLE II
TEST RESULTS

Load step (1250A/μs)	Amplitude ΔI	Spike ΔV /recovery time	
		w/ Aux.	w/o Aux.
0-~450A	~450A	50.0mV/5.4μs	74.3mV/5.4μs
300-50A	250A	71.0mV/7.2μs	111mV/8.1μs
50-300A	250A	30.5mV/5.9μs	40.7mV/6.0μs
250-50A	200A	48.8mV/5.7μs	76.8mV/7.0μs
300-150A	150A	38.1mV/4.6μs	43.1mV/4.9μs
0-~450A (w/o AVP)	~450A	41.2mV/8.2μs	68.1mV/8.7μs
300-50A (w/o AVP)	250A	59.6mV/6.9μs	113.7mV/9.1μs

The number of switches in the proposed auxiliary circuit will be 2 more compared to the previous solutions. However, due to the difference in the required drain-source breakdown voltage of the switches and the required drivers, the auxiliary circuit is more advantageous in terms of cost and area saving. For each additional voltage rail, the previous auxiliary circuit needs an additional pair of high-side and low-side MOSFETs with the drain-source breakdown voltage exceeding the input voltage V_g, which required a half-bridge driver to drive them. In contrast, the proposed auxiliary circuit only needs to add a pair of Back-to-Back MOSFETs whose drain-source breakdown voltage exceeds the output voltage V_{OUT}, and its driver only needs a pair of low-side drivers. Therefore, the more voltage rails, the greater the cost and space saving advantages of the proposed auxiliary circuit.

V. CONCLUSIONS

This paper proposes a practical auxiliary circuit that can suppress voltage spikes for multiple voltage rails and reduce the required output capacitance. The auxiliary circuit works independently and does not require any information from the VRM, making it easy to deploy without complex modifications to the original power supplies and minimizing wasteful duplication of updating VRM due to the increasing requirement of dynamic response. Significant cost and space savings can be achieved compared to the previous solutions, as the auxiliary circuit can be shared between several voltage rails. The improvement by the auxiliary circuit is analyzed in the time domain and verified. The full-scale prototype shows that the auxiliary circuit can suppress undershoot and overshoot by 25% and 36%, respectively, and are well adapted to functions such as AVP and DVFS in both voltage rails that are at different voltage levels. Besides VRM applications, the auxiliary circuit also suppresses the undershoot and overshoot by 40% and 48%, respectively, in PoL applications, which indicates this circuit is suitable for a wide range of power supply types.

TABLE III
COMPARISON OF THE PROPOSED AND EXISTING SOLUTIONS

	2020 TI [18]	2021 MPS [4]	2017 APEC [15]	2024 TPEL [14]	**This work**
V_g/V_{OUT}	12/1.0V	12/1V	12/1.8V	12/0.95V	**12/1V**
Phase num.	6	6	4+2 Aux.	6+2 Aux.	**6+1 Aux. (eq.)**
Aux. switch num.	-	-	4 (2N for N rails)	4 (2N for N rails)	**6 (2N+2 for N rails)**
L_n/L_a	150nH/ -	150nH/ -	300nH/ 150nH	150nH/ 70nH	**150nH/ 12nH(eq.)**
C_{OUT}	25mF	8.5mF	-	8.5mF	**6.3mF**
ΔI	225A	250A	240A	250A	**250A**
Slew rate	-	1200A/μs	800A/μs	1200A/μs	**1250A/μs**
f_{SW}/f_{SW_aux}	1MHz/ -	710kHz/ -	600kHz/ 600kHz	710kHz/ 1MHz	**1MHz/ -**
Undershoot/ recovery time	35mV/ 9μs	29mV/ 23μs	160mV/ 6μs	30mV/ 17μs	**30.5mV/ 5.9μs**
Overshoot/ recovery time	66mV/ 15μs	64mV/ 23μs	-	41mV/ 23μs	**71.0mV/ 7.2μs**
Support of multi-rails	-	-	No	No	**Yes**
FoM* (Step up/ down)[7]	0.028/ 0.009	0.057/ 0.025	-	0.081/ 0.044	**0.220/ 0.078**

*: $FoM = \frac{\text{load current step (A)}}{\text{output capacitance (mF)} \times \text{transient time } (\mu s) \times \text{switching frequency (MHz)} \times \text{voltage deviation (mV)}}$

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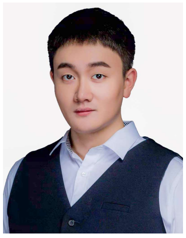

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